WORKSHEET 1B – MATHEMATICS

FORMAL ASSESSMENT | Questions and Answers | Total: 30 marks
21st Century Skills: Critical Thinking and Metacognition

Note: The worksheet asks about **burglary/fire** incidents, so every answer below is given for **both** data sets. Use **Option 1 (Number of Burglaries)** or **Option 2 (Number of Fires)** – whichever option you selected in Technology.

QUESTION 1: UNDERSTANDING THE DATA (4)

The table below shows the number of reported incidents of burglary and fire cases in South Africa over five years.

Year	Number of Burglaries (Option 1)	Number of Fires (Option 2)
2020	1 145	116
2021	1 981	215
2022	1 203	197
2023	1 579	256
2024	1 050	239
2025	**1 145 (predicted – Q3.2)**	**261 (predicted – Q3.2)**

1.1 In which year was the number of burglary/fire incidents the highest? (1)

A: Option 1 (burglaries): 2021 – 1 981 incidents.
Option 2 (fires): 2023 – 256 incidents.

1.2 In which year was the number of burglary/fire incidents the lowest? (1)

A: Option 1 (burglaries): 2024 – 1 050 incidents.
Option 2 (fires): 2020 – 116 incidents.

1.3 Calculate the range of burglary/fire incidents. (2)

A: Range = highest value – lowest value
Option 1: 1 981 – 1 050 = **931 incidents**
Option 2: 256 – 116 = **140 incidents**

QUESTION 2: MEASURES OF CENTRAL TENDENCY (7)

Using the data above (2020–2024):

2.1 Calculate the mean of burglary/fire incidents over the five years. (3)

A: Mean = sum of all values ÷ number of values
Option 1: (1 145 + 1 981 + 1 203 + 1 579 + 1 050) ÷ 5 = 6 958 ÷ 5 = **1 391,6 burglaries per year**
Option 2: (116 + 215 + 197 + 256 + 239) ÷ 5 = 1 023 ÷ 5 = **204,6 fires per year**

2.2 Determine the median of burglary/fire incidents over the five years. (2)

A: Arrange the values from smallest to largest and take the middle value.
Option 1: 1 050 ; 1 145 ; **1 203** ; 1 579 ; 1 981 ⇒ median = **1 203**
Option 2: 116 ; 197 ; **215** ; 239 ; 256 ⇒ median = **215**

2.3 Explain what the mean tells us about incidents of burglary/fire trends in South Africa over the five years. (2)

A: The mean gives the **average number of incidents per year** over the five years. It summarises the whole data set in one value, so it shows the general level of incidents and lets us compare any single year against a "typical" year. **Option 1:** with a mean of 1 391,6, the year 2021 (1 981) was far above average while 2024 (1 050) was well below it. **Option 2:** with a mean of 204,6, 2020 (116) was far below average while 2023 (256) was above it. The mean shows the overall level, but on its own it does not show the year-to-year trend.

QUESTION 3: PERCENTAGE INCREASE AND DECREASE (6)

3.1 Calculate the percentage decrease in burglary/fire incidents from 2023 to 2024. (3)

A: Percentage decrease = (old value – new value) ÷ old value × 100
Option 1: (1 579 – 1 050) ÷ 1 579 × 100 = 529 ÷ 1 579 × 100 = **33,50 % decrease**
Option 2: (256 – 239) ÷ 256 × 100 = 17 ÷ 256 × 100 = **6,64 % decrease**

3.2 The number of reported burglary or fire incidents increased by an average of 9 % from 2024 to 2025. Using the number of incidents recorded in 2024, calculate and predict the number of incidents for 2025. (3)

A: 2025 value = 2024 value × (100 % + 9 %) = 2024 value × 1,09
Option 1: 1 050 × 1,09 = 1 144,5 = **1 145 burglaries** (rounded to a whole incident)
Option 2: 239 × 1,09 = 260,51 = **261 fires** (rounded to a whole incident)

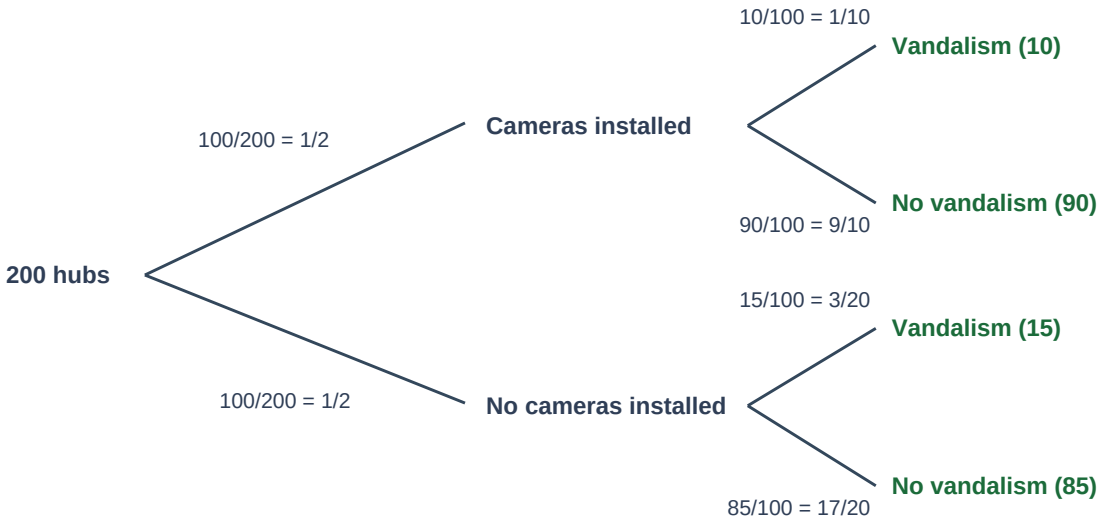
QUESTION 4: PROBABILITY AND RISK (5)

A survey also recorded whether community hubs had **security cameras** installed. Study the table below and answer the questions.

	Vandalism	No Vandalism	Total
Cameras Installed	10	90	100
No Cameras Installed	15	85	100
Total	**25**	**175**	**200**

4.1 Draw a tree diagram to represent the information in the table. Show the outcomes for a hub with cameras installed and without cameras installed, and whether vandalism occurred or did not occur. Label each branch with its probability as a fraction. (2)

A:



4.2 What is the probability that a randomly selected community hub has cameras and reported vandalism? (1)

A: P(cameras and vandalism) = 10 ÷ 200 = **1/20 = 0,05 = 5 %**

4.3 Determine the probability, as a percentage, that a community hub did not have cameras and experienced vandalism. (1)

A: P(no cameras and vandalism) = 15 ÷ 200 = 0,075 = **7,5 %**

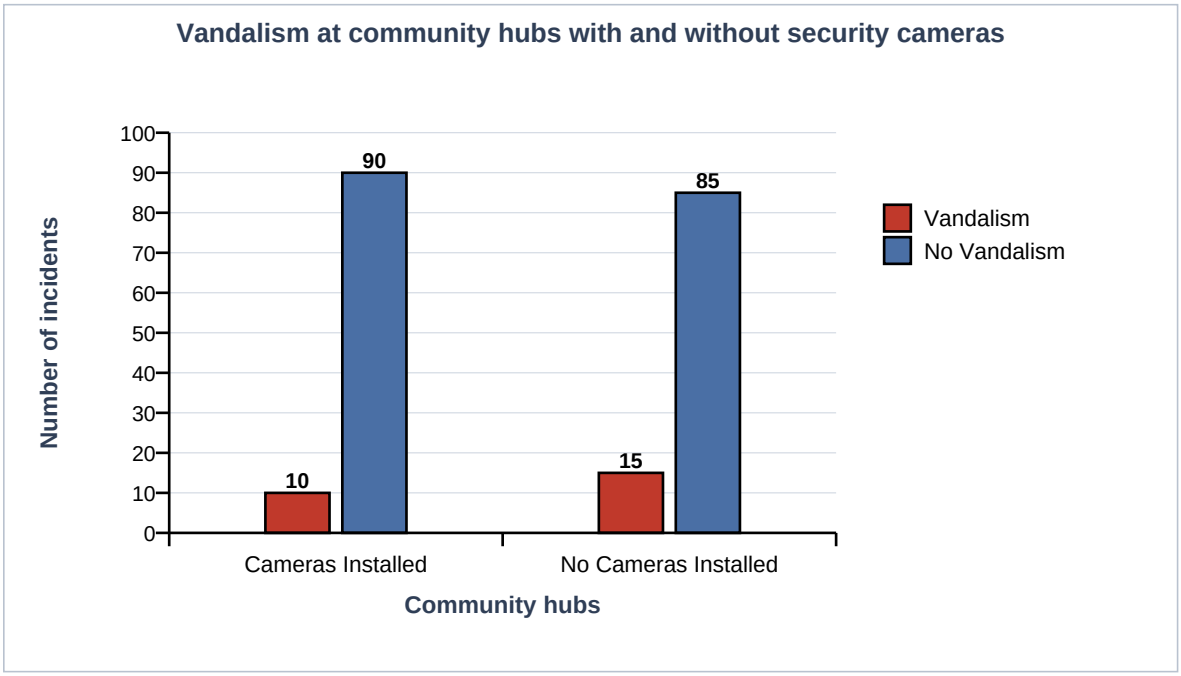
4.4 Which group of community hubs has a higher risk of vandalism? Justify your answer using the probabilities. (1)

A: **The hubs with no cameras installed have the higher risk.** Of the 100 hubs without cameras, 15 were vandalised (15/100 = 15 %), compared with only 10 of the 100 hubs with cameras (10/100 = 10 %). Security cameras therefore lower the risk of vandalism.

QUESTION 5: REPRESENTING DATA (8)

5. Use the information above to draw a double bar graph representing the information presented in Question 4. On the x-axis, indicate Cameras Installed and No Cameras Installed. On the y-axis, indicate Number of vandalism incidents. (8)

A: Marks are awarded for the **title**, both **axis labels**, a suitable **scale** (go up in 10s from 0 to 100), **correctly plotted bars**, a **key** and neat presentation. Bars to plot: Cameras Installed – vandalism 10, no vandalism 90; No Cameras Installed – vandalism 15, no vandalism 85.



Tip: if your teacher wants only the vandalism data, plot just the two red bars (10 and 15) and use a scale going up in 2s or 5s so the difference is easy to see.

TOTAL: 30 MARKS